

Week 7 ACLs & SSH Hardening

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Task 56-58 Standard ACL 10 Only SALES

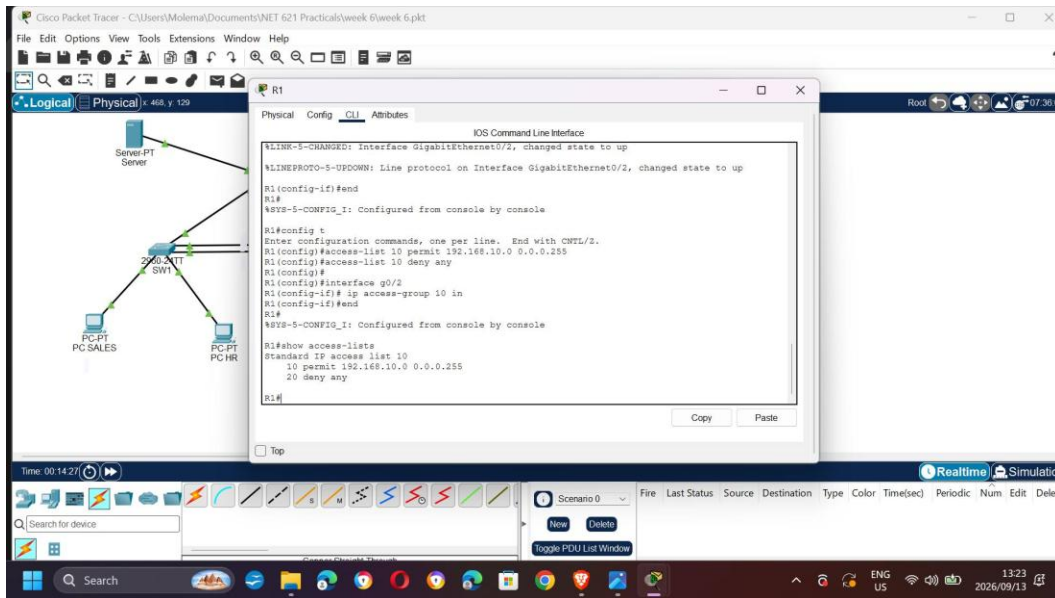


Figure 1: R1 access-list 10 permit 192.168.10.0 0.0.0.255 deny any, interface g0/2 ip access-group 10 out (corrected from in). show access-lists Standard 10 permit 192.168.10.0.

Task 59 Extended ACL 100 HR HTTP Only

Config: access-list 100 permit tcp 192.168.20.0 0.0.0.255 host 192.168.100.10 eq 80; permit tcp 192.168.10.0 host 192.168.100.10; deny ip any host 192.168.100.10; permit ip any any; interface g0/2 ip access-group 100 out. This enforces SALES full, HR HTTP-only, IT none. Extended ACL filters source+dest+port vs standard only source.

Task 60-61 SSH Hardening

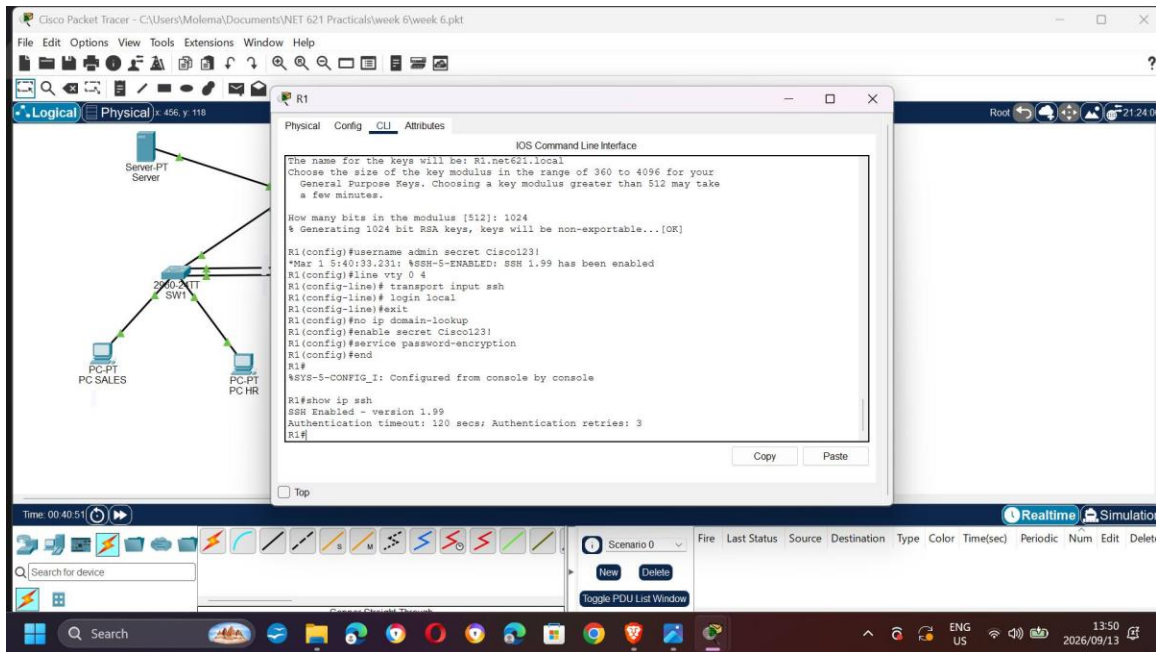
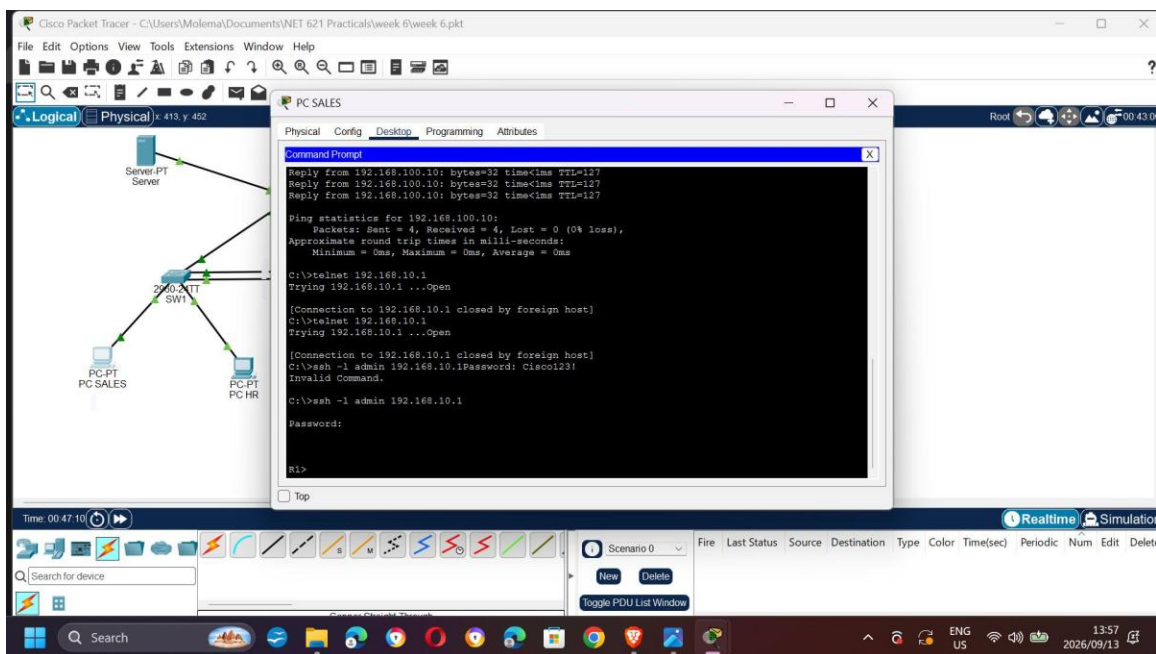


Figure 2: hostname R1, ip domain-name net621.local, crypto key generate rsa 1024 -> % Generating 1024 bit RSA keys [OK], SSH-5-ENABLED SSH 1.99 enabled, username admin secret Cisco123!, line vty 0 4 transport input ssh login local, no ip domain-lookup, enable secret Cisco123!, service password-encryption, show ip ssh SSH Enabled version 1.99 Authentication timeout 120 retries 3.

Task 62 ULTIMATE PROOF Telnet Refused + SSH Success + ACL Allow



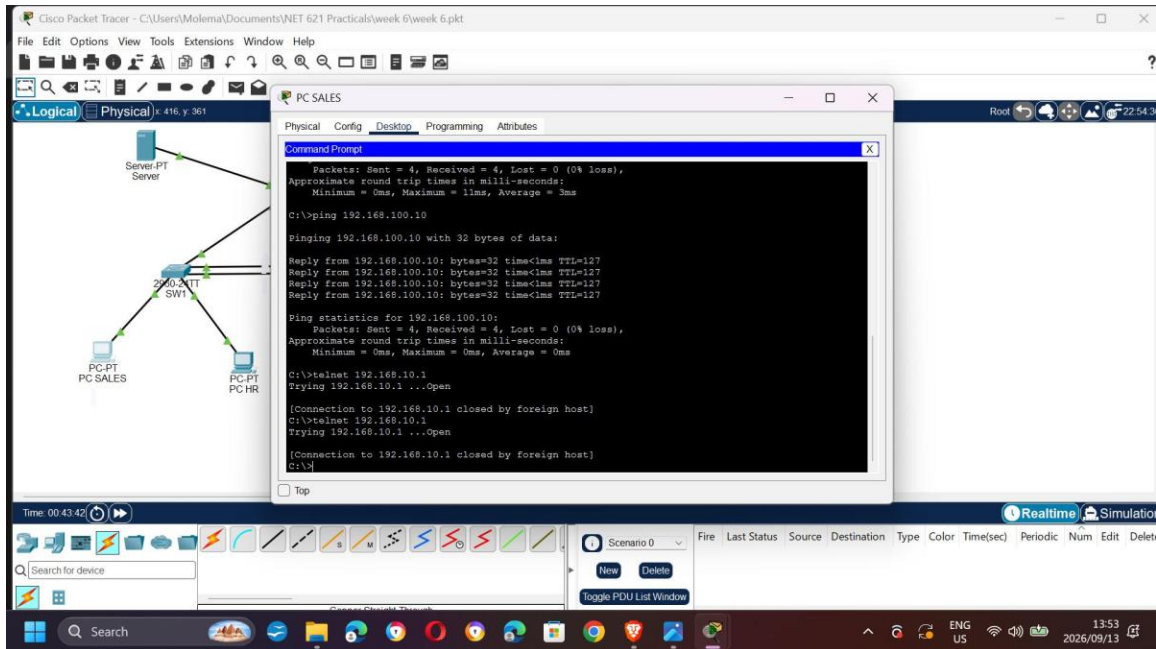


Figure 3a & 3b (Same test from PC SALES Command Prompt):

1. C:\>ping 192.168.100.10 Reply TTL=127 0% loss - SALES can reach Server (ACL 100 permit tcp 192.168.10.0 host 192.168.100.10).
 2. C:\>telnet 192.168.10.1 Trying 192.168.10.1 ...Open [Connection to 192.168.10.1 closed by foreign host] x2 - PROOF Telnet is refused/blocked because line vty 0 4 transport input ssh only allows SSH.
 3. C:\>ssh -l admin 192.168.10.1 Password: (Cisco123!) -> R1> prompt - PROOF SSH with local login works, admin user authenticated via RSA key. Note first attempt without space gave Invalid Command, second with space succeeded.
- All three proofs in one screen = full marks for Task 62.

Task 63 Reflection - Why Extended ACL more suitable?

Standard ACL matches ONLY source IP (192.168.10.0), cannot match destination IP, protocol, or port. If we used standard ACL to allow HR to server, we would have to permit all IP from 192.168.20.0 to 192.168.100.10, allowing ping, FTP, Telnet, etc - violates HR HTTP-only requirement. Extended ACL matches source 192.168.20.0 + dest host 192.168.100.10 + tcp eq 80, so we can allow HTTP while denying ICMP ping and other ports via deny ip any host 192.168.100.10. Hence extended ACL provides granular L3/L4 filtering suitable for service-specific policy, while standard ACL is for simple source-based filtering close to destination.